

```
In [1]: from IPython.display import Image
url = 'https://img.etimg.com/thumb/msid-71806721,width-650,imgsize-807917,,resizemo
Image(url,height=300,width=400)
```

Out[1]:



```
In [2]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
sns.set()
import warnings
data=pd.read_csv(r"C:\Users\Lenovo\Documents\Notes\APRIL\4th - REGRESSION PROJECT\R
data
```

Out[2]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8
1	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8
3	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5
4	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	6
...	...	...	...	...	...	...	...	...	...
18244	7	2018-02-04	1.63	17074.83	2046.96	1529.20	0.00	13498.67	13
18245	8	2018-01-28	1.71	13888.04	1191.70	3431.50	0.00	9264.84	8
18246	9	2018-01-21	1.87	13766.76	1191.92	2452.79	727.94	9394.11	9
18247	10	2018-01-14	1.93	16205.22	1527.63	2981.04	727.01	10969.54	10
18248	11	2018-01-07	1.62	17489.58	2894.77	2356.13	224.53	12014.15	11

18249 rows × 14 columns



In [3]: data.info()

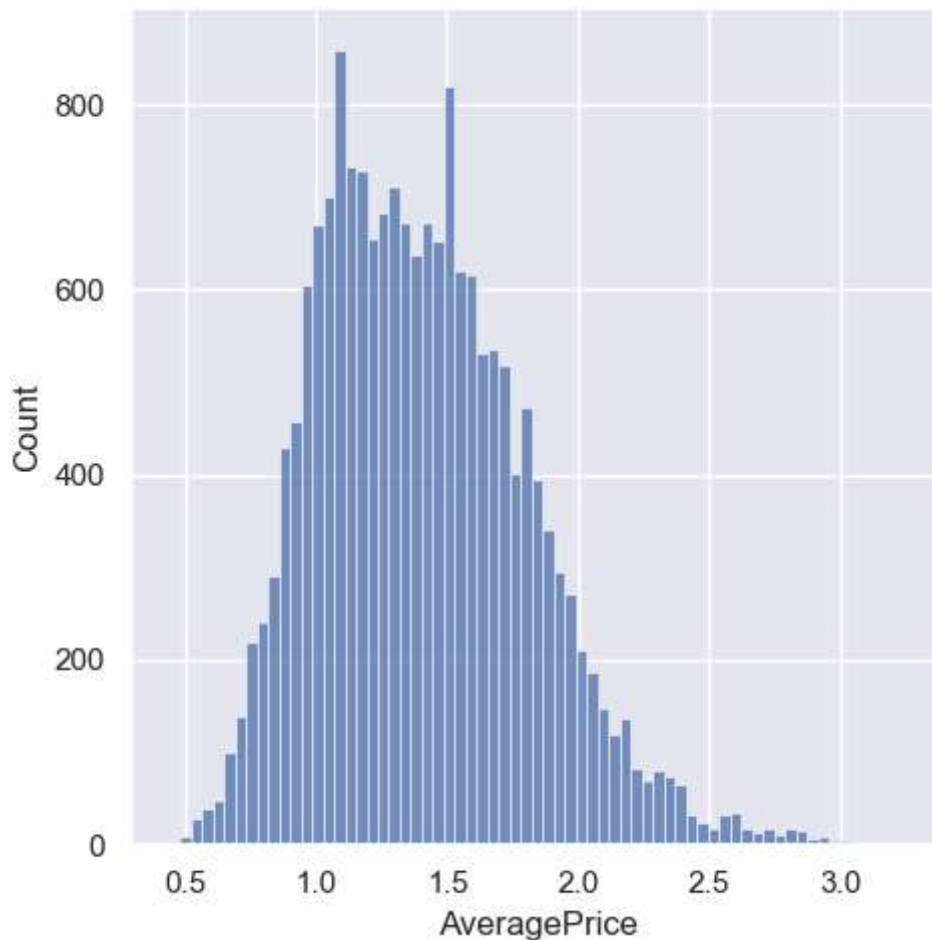
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 18249 entries, 0 to 18248
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Unnamed: 0            18249 non-null  int64
1   Date                  18249 non-null  object
2   AveragePrice          18249 non-null  float64
3   Total Volume         18249 non-null  float64
4   4046                  18249 non-null  float64
5   4225                  18249 non-null  float64
6   4770                  18249 non-null  float64
7   Total Bags           18249 non-null  float64
8   Small Bags           18249 non-null  float64
9   Large Bags           18249 non-null  float64
10  XLarge Bags          18249 non-null  float64
11  type                  18249 non-null  object
12  year                  18249 non-null  int64
13  region                18249 non-null  object
dtypes: float64(9), int64(2), object(3)
memory usage: 1.9+ MB
```

```
In [4]: data.head()
```

Out[4]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62
1	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9408.07
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21
3	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5677.40
4	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	5986.26

```
In [5]: sns.displot(data['AveragePrice']);
```



```
In [6]: sns.distplot(data['AveragePrice']);
```

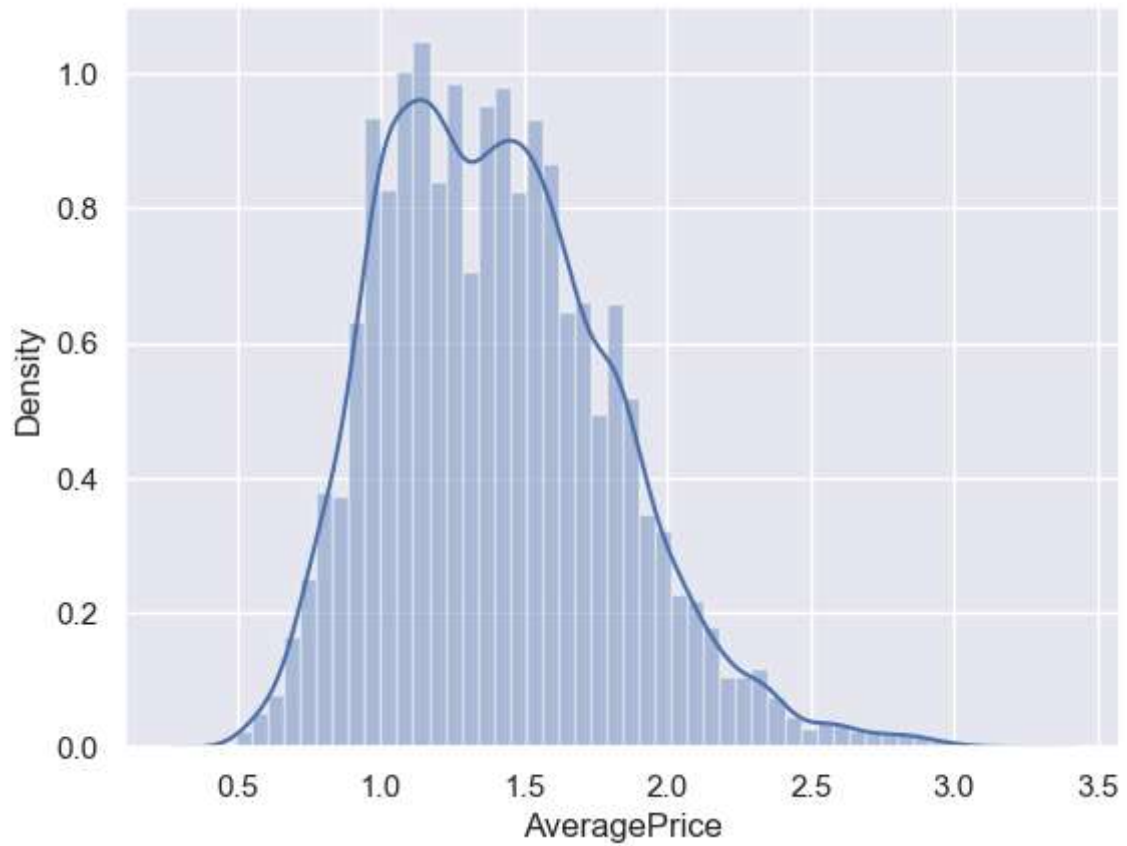
C:\Users\Lenovo\AppData\Local\Temp\ipykernel_960\2985105532.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

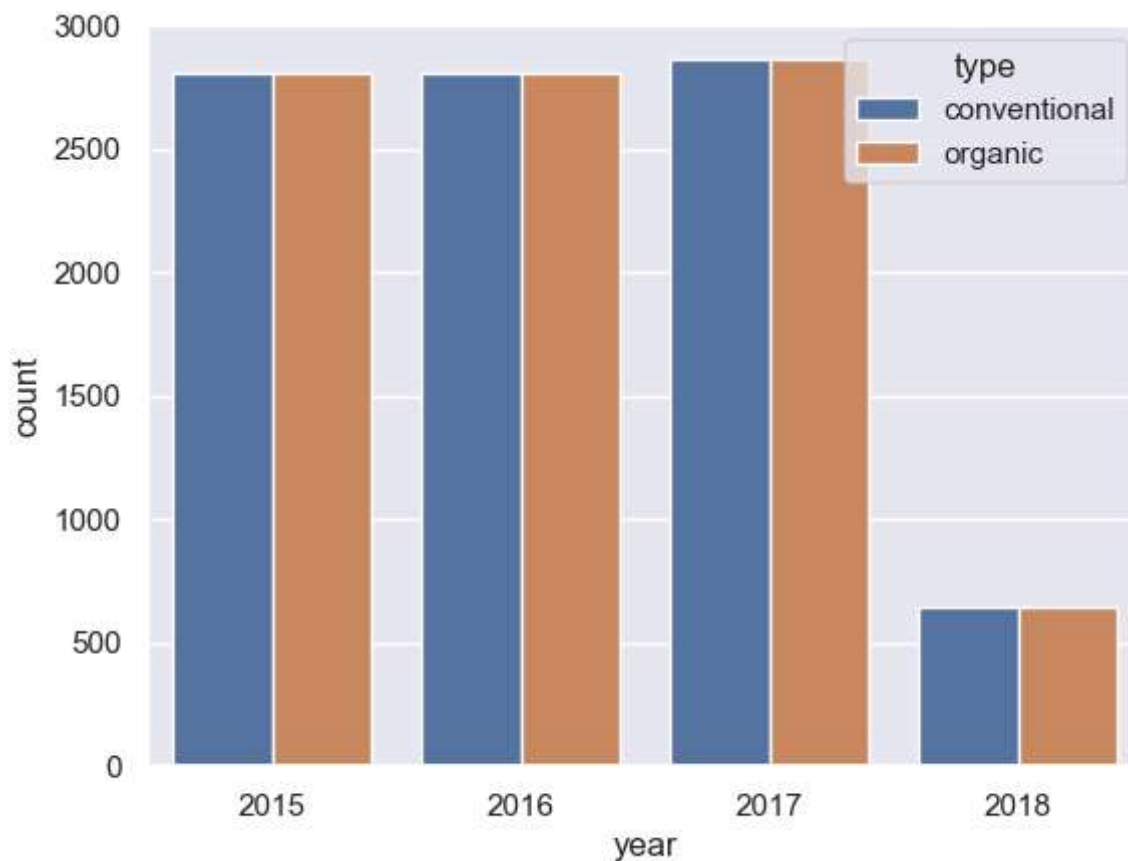
For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(data['AveragePrice']);
```



```
In [7]: sns.countplot(x='year',data=data,hue='type')
```

```
Out[7]: <Axes: xlabel='year', ylabel='count'>
```



```
In [8]: data.year.value_counts()
```

```
Out[8]: year
2017    5722
2016    5616
2015    5615
2018    1296
Name: count, dtype: int64
```

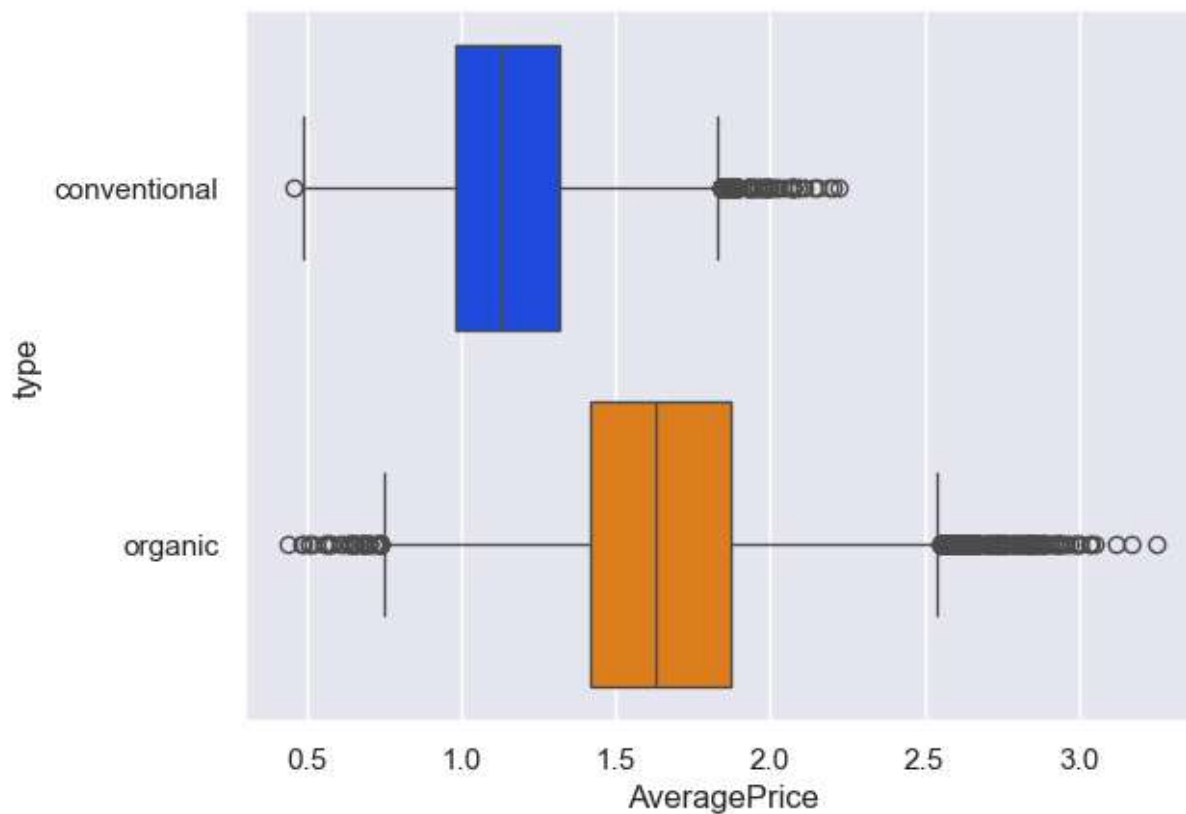
```
In [9]: sns.boxplot(y='type',x='AveragePrice',data=data,palette='bright')
```

C:\Users\Lenovo\AppData\Local\Temp\ipykernel_960\2303690611.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(y='type',x='AveragePrice',data=data,palette='bright')
```

```
Out[9]: <Axes: xlabel='AveragePrice', ylabel='type'>
```



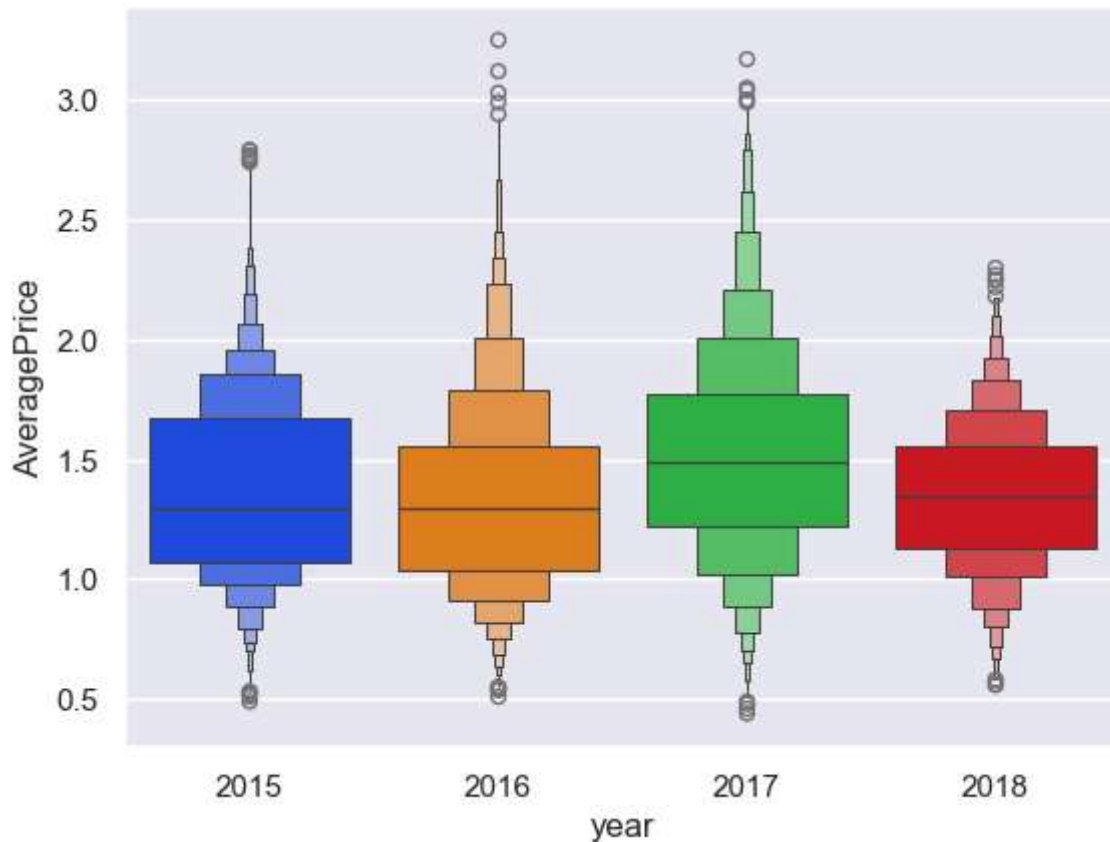
```
In [10]: data.year=data.year.apply(str)
sns.boxenplot(x='year',y='AveragePrice',data=data,palette='bright')
```

C:\Users\Lenovo\AppData\Local\Temp\ipykernel_960\3738379805.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxenplot(x='year',y='AveragePrice',data=data,palette='bright')
```

```
Out[10]: <Axes: xlabel='year', ylabel='AveragePrice'>
```



```
In [11]: data['type']=data['type'].map({'conventional':0,'organic':1})

#just taking month from date column

data.Date=data.Date.apply(pd.to_datetime)
data['Month']=data['Date'].apply(lambda x:x.month)
data.drop('Date',axis=1,inplace=True)
data.Month=data.Month.map({1:"JAN",2:'FEB',3:'MAR',4:'APR',5:'MAY',6:'JUN',7:'JUL',

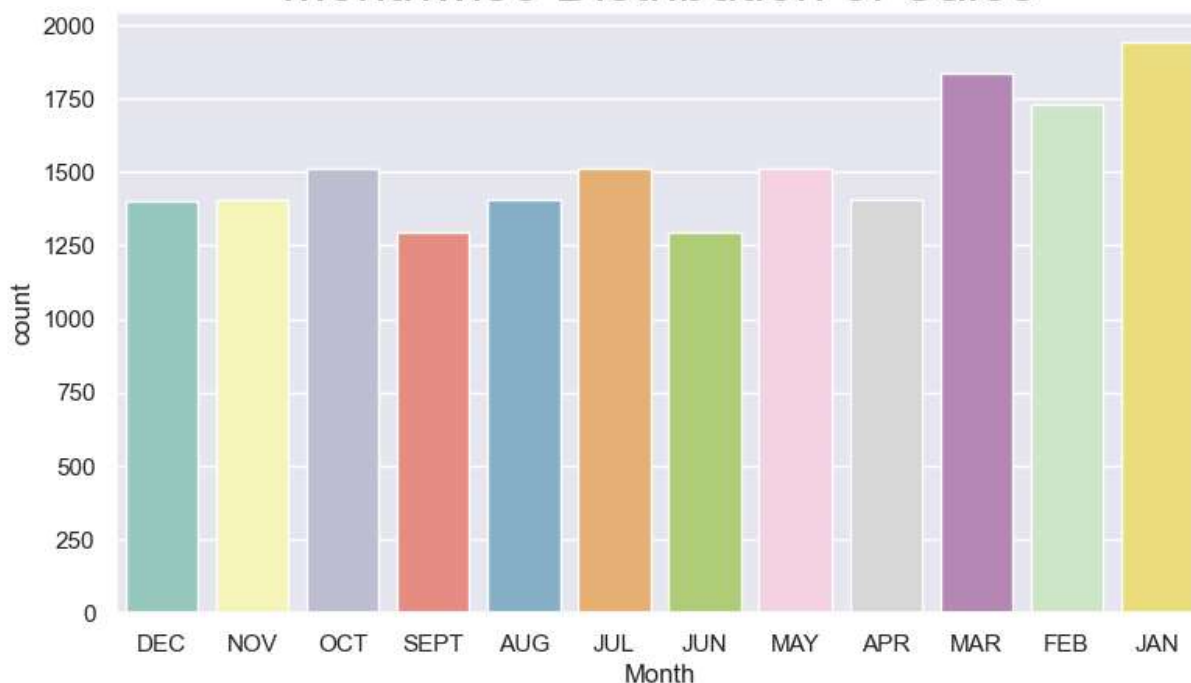
In [12]: plt.figure(figsize=(9,5))
#sns.countplot(data['Month'])
sns.countplot(x='Month',data=data,palette='Set3')
plt.title('Monthwise Distribution of Sales',fontdict={'fontsize':25});
```

C:\Users\Lenovo\AppData\Local\Temp\ipykernel_960\427997742.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.countplot(x='Month',data=data,palette='Set3')
```

Monthwise Distribution of Sales



```
In [13]: dummies=pd.get_dummies(data[['year','region','Month']],drop_first=True)
df_dummies=pd.concat([data[['Total Volume','4046','4225','4770','Total Bags',
                             'Small Bags','Large Bags','XLarge Bags','type']],dummi
target=data['AveragePrice']

#split train and test data
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(df_dummies,target,test_size=0.3)

#strandardize the data:

cols_to_std=['Total Volume','4046','4225','4770','Total Bags','Small Bags','XLarge

from sklearn.preprocessing import StandardScaler
scaler=StandardScaler()
scaler.fit(x_train[cols_to_std])
x_train[cols_to_std]=scaler.transform(x_train[cols_to_std])
x_test[cols_to_std]=scaler.transform(x_test[cols_to_std])
```

```
In [14]: from sklearn.linear_model import LinearRegression
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
from sklearn.svm import SVR
from sklearn.neighbors import KNeighborsRegressor
from sklearn.metrics import mean_absolute_error,mean_squared_error,r2_score
from xgboost import XGBRegressor
```

```
In [15]: regressors={
    'LinearRegression':LinearRegression(),
    'Decision Tree':DecisionTreeRegressor(),
    'Random Forest':RandomForestRegressor(),
    'Support Vector Machine':SVR(gamma=1),
```

```
'K-nearest Neighbors':KNeighborsRegressor(n_neighbors=1),  
'XGBoost':XGBRegressor()  
}  
results=pd.DataFrame(columns=['MAE',"MSE",'R2-score'])  
for method,func in regressors.items():  
    model =func.fit(x_train,y_train)  
    pred=model.predict(x_test)  
    results.loc[method]=[np.round(mean_absolute_error(y_test,pred),3),  
                        np.round(mean_squared_error(y_test,pred),3),  
                        np.round(r2_score(y_test,pred),3)]
```

```
In [24]: x_train,x_val,y_train,y_val=train_test_split(x_train,y_train,test_size=0.2)
```

```
In [27]: pip install tensorflow
```

Requirement already satisfied: tensorflow in c:\users\lenovo\anaconda3\lib\site-packages (2.19.0)

Requirement already satisfied: absl-py>=1.0.0 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (2.2.2)

Requirement already satisfied: astunparse>=1.6.0 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (1.6.3)

Requirement already satisfied: flatbuffers>=24.3.25 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (25.2.10)

Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (0.6.0)

Requirement already satisfied: google-pasta>=0.1.1 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (0.2.0)

Requirement already satisfied: libclang>=13.0.0 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (18.1.1)

Requirement already satisfied: opt-einsum>=2.3.2 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (3.4.0)

Requirement already satisfied: packaging in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (24.1)

Requirement already satisfied: protobuf!=4.21.0,!4.21.1,!4.21.2,!4.21.3,!4.21.4,!4.21.5,<6.0.0dev,>=3.20.3 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (5.29.4)

Requirement already satisfied: requests<3,>=2.21.0 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (2.32.3)

Requirement already satisfied: setuptools in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (75.1.0)

Requirement already satisfied: six>=1.12.0 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (1.16.0)

Requirement already satisfied: termcolor>=1.1.0 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (3.0.1)

Requirement already satisfied: typing-extensions>=3.6.6 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (4.12.2)

Requirement already satisfied: wrapt>=1.11.0 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (1.14.1)

Requirement already satisfied: grpcio<2.0,>=1.24.3 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (1.71.0)

Requirement already satisfied: tensorboard~2.19.0 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (2.19.0)

Requirement already satisfied: keras>=3.5.0 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (3.9.2)

Requirement already satisfied: numpy<2.2.0,>=1.26.0 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (1.26.4)

Requirement already satisfied: h5py>=3.11.0 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (3.11.0)

Requirement already satisfied: ml-dtypes<1.0.0,>=0.5.1 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorflow) (0.5.1)

Requirement already satisfied: wheel<1.0,>=0.23.0 in c:\users\lenovo\anaconda3\lib\site-packages (from astunparse>=1.6.0->tensorflow) (0.44.0)

Requirement already satisfied: rich in c:\users\lenovo\anaconda3\lib\site-packages (from keras>=3.5.0->tensorflow) (13.7.1)

Requirement already satisfied: namex in c:\users\lenovo\anaconda3\lib\site-packages (from keras>=3.5.0->tensorflow) (0.0.8)

Requirement already satisfied: optree in c:\users\lenovo\anaconda3\lib\site-packages (from keras>=3.5.0->tensorflow) (0.15.0)

Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\lenovo\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (3.3.2)

Requirement already satisfied: idna<4,>=2.5 in c:\users\lenovo\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (3.3.2)

ckages (from requests<3,>=2.21.0->tensorflow) (3.7)
 Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\lenovo\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (2.2.3)
 Requirement already satisfied: certifi>=2017.4.17 in c:\users\lenovo\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (2025.1.31)
 Requirement already satisfied: markdown>=2.6.8 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorboard~=2.19.0->tensorflow) (3.4.1)
 Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorboard~=2.19.0->tensorflow) (0.7.2)
 Requirement already satisfied: werkzeug>=1.0.1 in c:\users\lenovo\anaconda3\lib\site-packages (from tensorboard~=2.19.0->tensorflow) (3.0.3)
 Requirement already satisfied: MarkupSafe>=2.1.1 in c:\users\lenovo\anaconda3\lib\site-packages (from werkzeug>=1.0.1->tensorboard~=2.19.0->tensorflow) (2.1.3)
 Requirement already satisfied: markdown-it-py>=2.2.0 in c:\users\lenovo\anaconda3\lib\site-packages (from rich->keras>=3.5.0->tensorflow) (2.2.0)
 Requirement already satisfied: pygments<3.0.0,>=2.13.0 in c:\users\lenovo\anaconda3\lib\site-packages (from rich->keras>=3.5.0->tensorflow) (2.15.1)
 Requirement already satisfied: mdurl~=0.1 in c:\users\lenovo\anaconda3\lib\site-packages (from markdown-it-py>=2.2.0->rich->keras>=3.5.0->tensorflow) (0.1.0)
 Note: you may need to restart the kernel to use updated packages.

In [29]: `#x_train,x_val,y_train,y_val=train_test_split(x_train,y_train,test_size=0.2)`

```
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense,Activation,Dropout
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import EarlyStopping

model=Sequential()
model.add(Dense(76,activation='relu',kernel_initializer=tf.random_uniform_initializer(
    bias_initializer=tf.random_uniform_initializer(minval=0.1,maxval=0.1)))
model.add(Dense(200,activation='relu',kernel_initializer=tf.random_uniform_initializer(
    bias_initializer=tf.random_uniform_initializer(minval=-0.1)))
model.add(Dropout(0.5))
model.add(Dense(200,activation='relu',kernel_initializer=tf.random_uniform_initializer(
    bias_initializer=tf.random_uniform_initializer(minval=-0.1,maxval=0.1)))
model.add(Dropout(0.5))
model.add(Dense(200,activation='relu',kernel_initializer=tf.random_uniform_initializer(
    bias_initializer=tf.random_uniform_initializer(minval=-0.1, maxval=0.1)))
model.add(Dropout(0.5))
model.add(Dense(1))

model.compile(optimizer='Adam',loss='mean_squared_error')
early_stop=EarlyStopping(monitor='val_loss',mode='min',verbose=0,patience=10)
```

```

-----
ValueError                                Traceback (most recent call last)
Cell In[29], line 10
      7 from tensorflow.keras.callbacks import EarlyStopping
      9 model=Sequential()
----> 10 model.add(Dense(76,activation='relu',kernel_initializer=tf.random_uniform_ini
tializer(minval=-0.1,maxval=0.1),
      11                bias_initializer=tf.random_uniform_initializer(minval=0.1,ma
xval=0.1)))
      12 model.add(Dense(200,activation='relu',kernel_initializer=tf.random_uniform_i
nitializer(minval=-0.1,maxval=0.1),
      13                bais_initializer=tf.random_uniform_initailizer(minval=-0.
1)))
      14 model.add(Dropout(0.5))

File ~\anaconda3\Lib\site-packages\keras\src\layers\core\dense.py:87, in Dense.__ini
t__(self, units, activation, use_bias, kernel_initializer, bias_initializer, kernel_r
egularizer, bias_regularizer, activity_regularizer, kernel_constraint, bias_constra
int, lora_rank, **kwargs)
      72 def __init__(
      73     self,
      74     units,
      (...)
      85     **kwargs,
      86 ):
----> 87     super().__init__(activity_regularizer=activity_regularizer, **kwargs)
      88     self.units = units
      89     self.activation = activations.get(activation)

File ~\anaconda3\Lib\site-packages\keras\src\layers\layer.py:289, in Layer.__init__
(self, activity_regularizer, trainable, dtype, autocast, name, **kwargs)
      287     self._input_shape_arg = input_shape_arg
      288 if kwargs:
--> 289     raise ValueError(
      290         "Unrecognized keyword arguments "
      291         f"passed to {self.__class__.__name__}: {kwargs}"
      292     )
      294 self._path = None # Will be determined in `build_wrapper`
      295 self.built = False

ValueError: Unrecognized keyword arguments passed to Dense: {'kernel_intializer': <t
ensorflow.python.ops.init_ops_v2.RandomUniform object at 0x00000174E2DF6510>}

```

```

In [31]: model.fit(x=x_train.values,y=y_train.values,
                  validation_data=(x_val.values,y_val.values),
                  batch_size=100,epochs=150,callbacks=[early_stop])

```

```

-----
NameError                                Traceback (most recent call last)
Cell In[31], line 3
      1 model.fit(x=x_train.values,y=y_train.values,
      2            validation_data=(x_val.values,y_val.values),
----> 3            batch_size=100,epochs=150,callbacks=[early_stop])

NameError: name 'early_stop' is not defined

```

```
In [33]: losses=pd.DataFrame(model.history.history)
losses[['loss', 'val_loss']].plot()
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[33], line 1
----> 1 losses=pd.DataFrame(model.history.history)
      2 losses[['loss', 'val_loss']].plot()

AttributeError: 'Sequential' object has no attribute 'history'
```

```
In [35]: dnn_pred=model.predict(x_test)
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[35], line 1
----> 1 dnn_pred=model.predict(x_test)

File ~\anaconda3\Lib\site-packages\keras\src\utils\traceback_utils.py:122, in filter_traceback.<locals>.error_handler(*args, **kwargs)
    119     filtered_tb = _process_traceback_frames(e.__traceback__)
    120     # To get the full stack trace, call:
    121     # `keras.config.disable_traceback_filtering()`
--> 122     raise e.with_traceback(filtered_tb) from None
    123 finally:
    124     del filtered_tb

File ~\anaconda3\Lib\site-packages\keras\src\models\sequential.py:172, in Sequential.build(self, input_shape)
    170     return
    171 if not self._layers:
--> 172     raise ValueError(
    173         f"Sequential model {self.name} cannot be built because it has "
    174         "no layers. Call `model.add(layer)`."
    175     )
    176 if isinstance(self._layers[0], InputLayer):
    177     if self._layers[0].batch_shape != input_shape:

ValueError: Sequential model sequential cannot be built because it has no layers. Call `model.add(layer)`.
```

```
In [37]: results.loc['Deep Neural Network']=[mean_absolute_error(y_test,dnn_pred).round(3),mean_squared_error(y_test,dnn_pred).round(3),r2_score(y_test,dnn_pred).round(3)]
results
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[37], line 1
----> 1 results.loc['Deep Neural Network']=[mean_absolute_error(y_test,dnn_pred).round(3),mean_squared_error(y_test,dnn_pred).round(3),r2_score(y_test,dnn_pred).round(3)]
      2 results
      3 results

NameError: name 'dnn_pred' is not defined
```

```
In [39]: f"10% of mean of target variable is {np.round(0.1 * data.AveragePrice.mean(),3)}"
```

```
Out[39]: '10% of mean of target variable is 0.141'
```

```
In [41]: results.sort_values('R2-score',ascending=False).style.background_gradient(cmap='Gre
```

```
Out[41]:
```

	MAE	MSE	R2-score
XGBoost	0.093000	0.017000	0.899000
Random Forest	0.096000	0.019000	0.885000
Decision Tree	0.128000	0.039000	0.767000
LinearRegression	0.182000	0.058000	0.654000
Support Vector Machine	0.281000	0.129000	0.228000
K-nearest Neighbors	0.316000	0.182000	-0.087000

```
In [ ]:
```